

		Parul University Faculty of Engineering and Technology Parul Institute of Engineering and Technology AIML/ROBO AI /AI /AIDS /CYBER /MICRO /SAP/ QUICK/ ORACLE /BDA/ BCCCIOT /M&C	
Subject Name	Quant and Reasoning	A.Y.	2026-27
Subject Code	303105311	Semester	5th
Unit-1			
Sr No	Question	COs	B.T.
1	Find the smallest number divisible by both 21 and 5.	2	Apply
2	Evaluate $107 \times 107 + 93 \times 93$.	3	Evaluate
3	The H.C.F. of two numbers is 11 and their L.C.M. is 7700. If one of these numbers is 275 then find the other number.	2	Apply
4	$(2/5 \div 3/4) * 5/6$	2	Apply
5	$1164 \times 128 \div 8.008 + 969.007 = ?$	2	Apply
6	Is 5457 divisible by 7?	2	Apply
7	The number of prime numbers between 0 and 50 is _____.	2	Apply
8	What is the smallest natural number?	3	Evaluate
9	What is the HCF of 24 and 36?	2	Apply
10	Find the LCM of 8 and 12.	1	Remember
11	If the HCF of two numbers is 5 and their LCM is 60, then find the product of the two numbers.	2	Apply
12	A number divisible by both 2 and 3 is always divisible by _____.	2	Apply
13	Simplify: $15 + 25 \div 5 =$ _____.	3	Evaluate
15	Find the value of $3/4 + 1/4$		
16	What is the approximate value of 49×21 ?	3	Evaluate
17	The approximate value of $198 \div 10$ is _____.	2	Apply
18	The least number that is exactly divisible by 6, 8, and 12 is _____.	2	Apply
Unit-2			
Sr No	Question	COs	B.T
1	The first term of an AP is denoted by _____.	1	Remember
2	Sum of first 100 terms for the given A.P 2, 4, 6, 8, ... is	2	Apply
3	Which term of the A.P 10, 20, ... is 100?	2	Apply
4	If the first term of a GP is 2 and the common ratio is 3, then the fourth term is _____.	2	Apply
5	The average of runs of a cricket player of 10 innings was 32. How many runs must he make in his next innings so as to increase his average of runs by 4?	3	Apply
6	The average age of the group of 30 boys is 16 years. A boy of age 19 leaves the group and a new boy joins the group. If the new average age of the group is 15.8 years then find the age of the new boy.	3	Apply
7	Which term of the A.P. 5, 13, 21, ... is 181?	2	Apply
8	The average of 10, 20, and 30 is _____.	2	Apply

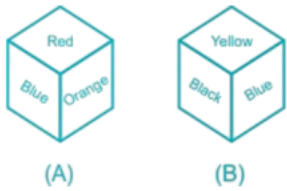
9	If the average of five numbers is 12, then what is their total sum?	2	Apply
10	What is the average of the first five natural numbers?	1	Remember
11	If the average of 8 observations is 15, then the sum of the observations is _____.	2	Apply
12	The average of 25 and 35 is _____.	2	Apply
13	In the Arithmetic Progression (AP) 2, 5, 8, 11, ..., then find common difference.	3	Evaluate
14	The 10th term of the AP 3, 6, 9, 12, ... is _____.	1	Apply
15	In the Geometric Progression (GP) 2, 6, 18, 54, ..., then what is common ratio?	2	Apply
16	The 5th term of the GP 2, 4, 8, 16, ... is _____.	2	Apply
17	What is the sum of the first 5 terms of the AP 1, 3, 5, 7, 9.	2	Apply
Unit-3			
Sr No	Question	COs	B.T
1	Ayesha's father was 38 years of age when she was born while her mother was 36 years old when her brother four years younger to her was born. What is the difference between the ages of her parents?	2	Apply
2	A father is twice as old as his daughter. If 20 years ago, the age of the father was 10 times the age of the daughter, what is the present age of the father?	3	Apply
3	If the price of a book is first decreased by 25% and then increased by 20%, then find the net change in the price.	2	Apply
4	If a father is 30 years older than his son and the sum of their ages is 50, then the age of the son is _____.	2	Apply
5	The ratio of boys to girls in a class is 3:2. If there are 15 boys, there are _____ girls.	2	Apply
6	Salaries of Ravi and Sumit are in the ratio 2:3. If the salary of each is increased by Rs. 4000, the new ratio becomes 40:57. What is Sumit's salary?	2	Apply
7	If $A:B = 7:5$ and $B:C = 9:11$, then $A:B:C$ is equal to _____.	2	Apply
8	A mother is 24 years older than her son. The sum of their present ages is 48 years. Find the present age of the son.	2	Apply
9	The price of a laptop is increased by 15% and then decreased by 10%. Find the net percentage change in the price.	2	Apply
10	The ratio of the ages of Ram and Shyam is 4:5. If Ram is 20 years old, find Shyam's age.	2	Apply
11	Salaries of Amit and Mohit are in the ratio 5:7. If Amit's salary is ₹25,000, find Mohit's salary.	2	Apply
12	If $P:Q = 3:4$ and $Q:R = 8:5$, then find the ratio $P:Q:R$.	2	Apply
Unit-4			
Sr No	Question	COs	B.T
1	At what rate of compound interest per annum will a sum of Rs. 1200 become Rs. 1348.32 in 2 years?	2	Apply
2	The percentage profit earned by selling an article for Rs. 1920 is equal to the percentage loss incurred by selling	2	Apply

	the same article for Rs. 1280. At what price should the article be sold to make 25% profit?		
3	A sum of money was invested at Simple Interest at a certain rate for 3 years. Had it been invested at a 4% higher rate. It would have fetched Rs 480 more. Find the Principal? What is the difference between Simple Interest and Compound Interest?	2	Apply
4	The sum of two numbers is 24 and their product is 143. Find the sum of their squares.	3	Apply
5	A and B start a business by investing Rs. 2000 and Rs. 3000 respectively. What is the ratio of their profits?	3	Apply
6	A and B enter into a partnership with Rs.50000 and Rs.75000 respectively in a company for a year. After 7 months, C gets into a partnership with them with Rs.30000 and A withdraws his contribution after 9 months. How would they share their profit of Rs.2600 at the end of the year?	2	Apply
7	A dishonest shopkeeper professes to sell cloth at the cost price, but he uses faulty meter rod. His meter rod measures 95 cm only. Find his gain per cent.	2	Apply
8	A sum of Rs. 5000 is invested at compound interest for 2 years at 10% per annum. Find the amount and the compound interest earned.	2	Apply
9	By selling 45 lemons for Rs. 40, a man loses 20%. How many lemons should he sell for Rs. 24 to gain 20% in the transaction?	2	Apply
10	A and B enter into a partnership. A invests Rs. 16000 for 8 months and B remains in the partnership for the full year. At the end of the year the profit is divided in the ratio 2:3. How much did B invest?	3	Apply
11	The difference between compound interest and simple interest on a certain sum at 10% per annum for 2 years is Rs. 631. Find the principal amount.	2	Apply
12	A shopkeeper marks his goods 20% above cost price and then gives a discount of 10%. Find his net gain percent.	2	Apply
13	The product of two numbers is 4107. If the HCF of these numbers is 37, find the greater number given that their sum is 185.	3	Apply
14	A, B, and C start a business with investments of Rs. 12000, Rs. 16000, and Rs. 20000 respectively. After 6 months, A doubles his investment and C withdraws Rs. 5000. If the total profit at the end of the year is Rs. 43200, find each partner's share.	2	Apply
15	A dishonest dealer claims to sell goods at cost price but uses a weight of 900 grams instead of 1 kg. Find his actual profit percentage. Also find what percentage above cost price he should mark the goods to earn a true profit of 15%.	2	Apply
Unit-5			

Sr No	Question	COs	B.T
1	The speed of the boat in still water is 5 km/hr. If the speed of the boat against the stream is 3 km/hr, what is the speed of the stream?	3	Apply
2	Two pipes A and B can fill a tank in 20 and 30 minutes respectively. If both pipes are opened together, how long will it take to fill the tank?	3	Apply
3	A boat covers a distance downstream in 1 hour, while it comes back in $1\frac{1}{2}$ hours. If the speed of the stream is 3 kmph, what is the speed of the boat in still water?	3	Apply
4	A can do a piece of work in 20 days. He works at it for 5 days and then B finishes it in 10 days. How long will they take to complete the work if they work together?	3	Apply
5	If a man covers 10.2 km in 3 hours then find the distance covered by him in 5 hours.	3	Apply
6	A alone can do a piece of work in 6 days and B alone in 8 days. A and B undertook to do it for Rs. 3200. With the help of C, they completed the work in 3 days. How much is to be paid to C?	3	Apply
7	In one hour, a boat goes 13 km/hr in the direction of the stream and 7 km/hr against the direction of the stream. What will be the speed of the boat in still water?	3	Apply
8	A person travels from one place to another at 30 km/hr and returns at 120 km/hr. If the total time taken is 5 hours, then find the Distance	3	Apply
9	Anita and Veena are running in opposite directions. Speed of Anita and Veena is 8 km/hr and 10 km/hr respectively. Find out the distance between them after 2.5 hours.	3	Apply
10	A train is moving with a speed of 92.4 km/hr. How many meters will it cover in 10 minutes?	3	Apply
11	Three pipes A, B, and C can fill a tank in 12, 15, and 20 minutes respectively. If all three pipes are opened together, in how much time will the tank be filled?	3	Apply
12	A motorboat whose speed is 18 km/hr in still water takes 1 hour more to go 24 km upstream than to return downstream to the same spot. Find the speed of the stream.	3	Apply
13	A and B together can complete a piece of work in 12 days. B and C together can complete the same work in 16 days. A and C together can complete it in 24 days. In how many days can A, B, and C together complete the work?	3	Apply
14	A train 150 m long passes a pole in 15 seconds and another train of the same length traveling in the opposite direction in 12 seconds. Find the speed of the second train.	3	Apply
15	Pipe A can fill a tank in 40 minutes and Pipe B can empty the full tank in 60 minutes. If both pipes are opened simultaneously, how long will it take to fill an empty tank completely?	3	Apply
Unit-6			

Sr No	Question	COs	B.T
1	Find the probability of getting a sum of 7 in a simultaneous throw of two dice.	2	Apply
2	From a group of 7 men and 6 women, five persons are to be selected to form a committee so that at least 3 men are there on the committee. In how many ways can it be done?	3	Apply
3	Find the number of permutations and combinations if $n = 12$ and $r = 2$.	1	Apply
4	Two dice are rolled, find the probability that the sum is: A. equal to 1 B. equal to 4	1	Apply
5	In how many ways a committee consisting of 5 men and 3 women, can be chosen from 9 men and 12 women?	2	Apply
6	What is the probability of getting two tails when two coins are tossed?	2	Apply
7	From a group of 7 men and 6 women, five persons are to be selected to form a committee so that at least 3 men are there on the committee. In how many ways can it be done?	3	Apply
8	A bag contains 5 red balls, 4 blue balls and 3 green balls. If 3 balls are drawn at random, find the probability that: A. all 3 are red B. exactly 2 are blue.	2	Apply
9	Find the number of ways in which the letters of the word 'EQUATION' can be arranged so that all the vowels always come together.	3	Apply
10	A card is drawn at random from a well-shuffled deck of 52 cards. Find the probability that the card drawn is: A. a king B. a red card C. a face card.	2	Apply
Unit-7			
Sr No	Question	COs	B.T
1	Siva, Sathish, Amar and Praveen are playing cards. Amar is to the right of Sathish, who is to the right of Siva. Who is to the right of Amar?	3	Apply
2	Rasik walked 20 m towards the north. Then he turned right and walked 30m. Then he turns right and walks 35m. Then he turns left and walks 15 m. Finally he turns left and walks 15 m. In which direction and how many meters is he from the starting position?	2	Apply
3	One morning after sunrise, Nandita and Ravi were sitting in a lawn with their backs towards each other. Nandita's shadow fell exactly towards her left-hand side. Which direction was Ravi facing?	3	Analyze
4	J,K,L,M,N,O,P and R are eight huts. L is 2 km east of K. J is 1 km north of K and Q is 2 km south of J. P is 1 km west of Q while M is 3 km east of P and O is 2 km north of P. R is situated right in the middle of K and L while N is just in the middle of Q and M. A. Distance between K and P is _____ B. Distance between K and R is _____	3	Analyze

5	Mr Deepak Mohan walks 5 km towards the south and then turns to the right. After walking 3 km he turns to the left and walks 5 km. What direction is he facing right now?	3	Analyze
6	In a Coaching Institute, seven students (both girls and boys) sit on three different benches, Bench @, Bench * and Bench #. Each bench must have at least two students sitting on it. At Least one girl must be sitting on each bench X, who is a girl, does not sit with Y, Z and T. S is a boy student and sits only with M. Y sits on the Bench @ with his best friends. H sits on Bench #. Z is the brother of X. A. Which of the following pair of students form the correct set of students sitting on the bench * B. H sits on Bench # along with _____ C. M sits along with S on which bench? D. Of the three benched @, * and #, which bench has 3 students sitting on it? E. Which of the given sets represents the students sitting on Bench @?	3	Analyze
7	Mohan starts from point A, walks 6 km towards the west, then turns right and walks 4 km, then turns right and walks 6 km, and finally turns right and walks 2 km to reach point B. In which direction is point B from point A, and what is the distance between them?	2	Apply
8	Five friends Ajay, Bina, Chetan, Divya and Esha are standing in a straight line facing south. Bina is to the immediate left of Esha. Ajay is to the immediate right of Chetan. Divya is at one of the extreme ends. Esha is second from the left end. A. Who is standing at the extreme right? B. Who is standing in the middle? C. Who are the immediate neighbours of Esha?	3	Analyze
9	One afternoon after noon, Ramesh and Suresh were standing back to back. Ramesh's shadow was falling exactly to his right. Which direction was Suresh facing?	3	Analyze
10	Eight persons A, B, C, D, E, F, G and H are sitting around a circular table facing the centre. B is second to the left of E. D is third to the right of B. A is second to the right of D. C sits between E and G. F is not an immediate neighbour of D. A. Who is sitting to the immediate right of E? B. Who is sitting opposite to D? C. Which pair of persons are immediate neighbours of A?	3	Analyze
Unit-8			
Sr No	Question	COs	B.T
1	Minute hand moves °(degrees) in 5 min.	2	Evaluate
2	At what time between 3 and 4 o'clock, the hands make an angle of 10 degrees?	3	Analyze
3	What is the angle between the two hands of a clock when the time shown by the clock is 6.30 p.m. ?	2	Apply
4	Today is Monday. After 61 days, it will be:	2	Analyze

5	What was the day of the week on 17th June, 1998?	2	Analyze
6	If 17th march 2008 was Monday, what was 1st April 2012?	2	Analyze
7	Which year in the future will have the same calendar exactly as 2009?	2	Analyze
8	At what time between 4 and 5 o'clock will the hands of a clock be together (i.e., coincide)?	3	Analyze
9	What is the angle between the hour hand and the minute hand of a clock at 3:40 p.m.? Also state which hand is ahead.	2	Apply
10	What was the day of the week on 15th August, 1947? Also, if a person was born on that day, what day of the week will their 75th birthday fall on?	2	Analyze
Unit-9			
Sr No	Question	COs	B.T
1	Four usual dice are thrown on the ground. The total of numbers on the top faces of these four dice is 13 as the top faces showed 4, 3, 1 and 5 respectively. What is the total of the faces touching the ground?	3	Apply
2	What number will be opposite to 2? 	3	Analyze
3	What is the opposite face of "Red"?  (A) (B)	3	Analyze
4	Choose the box that is similar to the box formed from the given sheet of paper (X).  (X) (1) (2) (3) (4)	3	Analyze
5	What was the day of the week on 17th June, 1998?	2	Analyze
6	If 17th march 2008 was Monday, what was 1st April 2012?	2	Analyze
7	Which year in the future will have the same calendar exactly as 2009?	2	Analyze
8	At what time between 4 and 5 o'clock will the hands of a clock be together (i.e., coincide)?	3	Analyze
9	What is the angle between the hour hand and the minute hand of a clock at 3:40 p.m.? Also state which hand is ahead.	2	Apply

10	What was the day of the week on 15th August, 1947? Also, if a person was born on that day, what day of the week will their 75th birthday fall on?	2	Analyze
Unit-10			
Sr No	Question	COs	B.T
1	If A is the brother of B; B is the sister of C; and C is the father of D, how is D related to A?	3	Analyze
2	If D is the brother of B, how is B related to C? To answer this question, which of the statements is/are necessary? A. The son of D is the grandson of C. B. B is the sister of D.	3	Analyze
3	Anupam said to a lady sitting in a car, "The only daughter of the brother of my wife is the sister-in-law of the brother of your sister." How is the husband of the lady related to Anupam?	3	Analyze
4	If A is the sister of B, B is the daughter of C, C is the wife of D, and D is the brother of E, how is E related to C?	3	Analyze
5	If A + B means A is the mother of B; A - B means A is the brother B; A % B means A is the father of B and A x B means A is the sister of B, which of the following shows that P is the maternal uncle of Q? A. Q - N + M x P B. P + S x N - Q C. P - M + N x Q. D. Q - S % P	2	Analyze
6	If A + B means A is the brother of B; A - B means A is the sister of B and A x B means A is the father of B. Which of the following means that C is the son of M? A. M - N x C + F B. F - C + N x M C. N + M - F x C D. M x N - C + F	3	Analyze
7	If X + Y means X is the daughter of Y; X - Y means X is the brother of Y; X % Y means X is the father of Y and X x Y means X is the sister of Y. Which of the following means I is the niece of J? A. J - N % C x I B. I x C - N % J C. J + M x C % I D. I x C + N - J	3	Analyze
8	Pointing to a photograph, Rahul said, "She is the daughter of the only son of my grandfather." How is the girl in the photograph related to Rahul? Also, if Rahul has a brother named Suresh, how is the girl related to Suresh?	3	Analyze
9	If A \$ B means A is the husband of B; A @ B means A is the daughter of B; A # B means A is the son of B and A * B means A is the mother of B. Which of the following means that R is the paternal grandmother of T? A. R * P \$ Q @ T B. T # P \$ Q * R C. R * P \$ Q # T D. T @ P * Q # R	3	Analyze
10	Introducing a man, a woman said, "His mother is the only daughter of my mother." How is the woman related to the man? If the man has a sister, how is the woman related to that sister?	3	Analyze
Unit-11			

Sr No	Question	COs	B.T
1	Look at this series: 664, 332, 340, 170, ____, 89, ... What number should fill the blank?	2	Apply
2	Look at this series: 72, 76, 73, 77, 74, __, 75, ... What number should fill the blank?	3	Apply
3	Look at this series: 53, 53, 40, 40, 27, 27, ... What number should come next?	1	Apply
4	Find the odd man out: 396, 462, 572, 427, 671, 264	2	Apply
5	Insert the missing number: 8, 7, 11, 12, 14, 17, 17, 22, (....)	2	Apply
6	If CALM is written as LKZB, YEAR may be written as	3	Evaluate
7	If 'MEANDER' is coded as '4515459', then 'MATHEMATICS' is coded as	3	Evaluate
8	Look at this series: 2, 6, 18, 54, ____, 486, ... What number should fill the blank? Also find the odd one out from the following: 3, 9, 27, 54, 81.	2	Apply
9	Insert the missing number: 3, 5, 9, 15, 23, 33, (____). Also find the odd man out: 121, 169, 196, 225, 251.	2	Apply
10	If SCHOOL is coded as FPKLLA, then COLLEGE may be coded as Also, if 'FLOWER' is coded as '615231823', then 'GARDEN' is coded as	3	Evaluate